

Report to DPP PV Task Force:

Portfolio edition: minor spelling, formatting, and privacy cleanup only. Content reflects the original date and context.

Responding to Flood Plain & Solar Installation Inquiries

Background

Throughout 2023, The City & County of Honolulu's (CCH) Department of Planning & Permitting (DPP) has been host to the DPP/PV Solar Task Force, meeting regularly to identify & address areas for improvement and compromise in the permitting process for customer-sited solar energy projects. The goal of the group is to clarify permit requirements, update existing regulations, expedite permit issuance, and reduce the backlog of DPP permit reviews. One area requiring clarification has been the interpretation and application of the FEMA Flood Zone's Substantial Improvement requirements. On November 22, 2023, HSEO met with partners from the Department of Land and Natural Resources, Federal Emergency Management Agency, and the Department of Planning & Permitting for interagency coordination on the subject. While renewable energy is not exempt from these improvement calculations, there is nuance to the interpretation of the FEMA guidelines. This document seeks to offer clarity on FEMA Special Flood Hazard Area (SFHA) requirements as they pertain to solar as well as identify opportunities based on the approaches from other Authorities Having Jurisdiction (AHJ).

SFHAs & Substantial Improvement Calculations:

When improvements are made to a structure within a high-risk flood zone (Zones beginning with the letters A or V; zones with a greater than 1% annual chance of flooding), they are subject to the Substantial Improvement / Substantial Damage (SI/SD) Calculation. By FEMA Regulation, if the cost of repairing or improving a building in an SFHA equals or exceeds 50 percent of the building's market value (excluding land value), then the project is deemed SI/SD and must first be made code compliant. Generally, this increased compliance requires having the lowest floor (including basement) elevated to above the base flood level. For the purposes of this group we will focus on Substantial Improvements rather than Substantial Damage calculations.

Substantial Improvement means any reconstruction, rehabilitation, addition, or other improvement of a structure, the cost of which equals or exceeds 50 percent of the market value of the structure before the "start of construction" of the improvement.

Differentiating FEMA Requirements & County Requirements

Under FEMA regulation, the Substantial Improvement calculation is project specific and does not require a multi-year calculation, however, to avoid a multi-year phased substantial improvement to a non-compliant flood vulnerable property, many jurisdictions apply the calculation to a multi-year improvement accumulation. A cumulative calculation is also included in Honolulu's recently adopted Community Rating System (CRS), which under Activity 430 credits Higher Regulatory Standards. Under ROH § 21A-1.4 City & County of Honolulu defines Substantial Improvements as those made over any 5-year period which exceed 50% of the value of the property at the start of the first improvement. For the counties of Kaua'i & Maui, this is a ten-year lookback. For Hawai'i County, it is three years. The cumulative calculation approach prevents overdevelopment of flood zones but is taxing on record keeping at the county level.

Responsibility to Calculate:

DPP's SI calculation is tedious. First, the county must verify the SFHA status of a structure as the flood zones are occasionally updated with new information. Then, they evaluate the cost of improvement provided, the cost of past improvements within a five-year window, and the value of the home less the land value to determine if the improvement is considered a substantial improvement. Finally, if the project is a SI, the county must determine if the structure is conforming or non-conforming based on the specified SFHA's development standards, and potentially the increased compliance

encouraged by the Community Rating System. These verifications are done prior to permit approval or issuance, meaning this could be time spent in review of an application that will not be approved for a variety of other reasons.

At a Task Force Meeting on June 13th, 2023, the County suggested that these calculations could potentially be done by the applicants and included upon submission to ease the workload on DPP.

FEMA regulations are not prescriptive of who performs an SI calculation, only that it must be verified by the AHJ over permitting. Included is a sample checklist for another AHJ (Hillsborough County, Florida), which requires calculations for these improvements to be done by the applicant in accordance with their local flood management ordinance.

To maintain consistency of results, the resources used by CCH DPP currently are:

Property values:

CCH Real Property Assessments

Other methods of calculation may be substituted as well

Cost of Improvements

CCH Open Data (not confirmed as reliable yet)

Lowest Floor Elevation

Elevation Certificates

Addressing Questions Posed by Task Force Members:

On September 18th, 2023, Task Force Members suggested that if an install is above the flood line, it should not be included in SI calculations and that similar approaches may have been taken in other jurisdictions.

SFHA classification for a structure is determined by the lowest floor of the structure relative to the Base Flood Elevation (BFE) for the SFHA. If the lowest floor is above the BFE the property is not subject to SI calculations. If it is not, then ALL specified improvements are subject to SI calculations. If solar is placed on the roof of a SFHA designated property, it must be included in the calculation as it is increasing the value of a structure at increased risk of flooding damage. One departure is if the BFE has been raised due to a map revision. Then only the improvements would need to be above the BFE.

On October 6th, 2023, DPP Requested HSEO assistance to find answers to the following questions:

What is the latest FEMA guideline regarding whether solar PV equipment does count towards the 5 year period for 50% improvements for non-conforming structures?

Renewable energy is not exempt from these improvement calculations, as dictated by FEMA regulation.

Is there a way to easily verify whether the structure/house is in the flood zone or just a portion of the land is in the flood zone so the 50% cost of improvements doesn't apply?

The SFHA classification is based on the lowest level of the bottom floor of the structure, rather than the elevation of the land. If any portion of the structure is below the BFE it must be brought into compliance with SFHA specifications.

Available Tools to Identify Floodplain BFE Include:

FEMA Flood Map Service Center

FEMA National Flood Hazard Layer Viewer

The above can be compared to an Elevation Certificate on file at DPP or included in the application.

Is there an easy way to verify whether the existing structure already complies with the flood zone requirements and is not subject to the 50% cost of improvements rule?

FEMA has specified regulations that can differ depending on the specific flood zone. However, per the FEMA Substantial Improvement and Substantial Damage Desk Reference, "The most common floodplain management requirement that exceeds NFIP minimums and also applies to SI/SD is freeboard – added height to raise the lowest floor above the BFE." Therefore, if the lowest floor of the home has been elevated, it would be compliant by FEMA standards, but still would need to meet any other Honolulu specific conditions, including the CRS, which may have a higher BFE requirement than FEMA.

Can I build a PV Installation that is detached from my home or structure? Would that be exempt from SI calculations?

If freestanding solar/photovoltaic panels are outside of the footprint of a structure and still in the floodplain, they are classified as a development and require permits. The local officials must determine if they are reasonably safe from flooding, including if they are properly anchored, raised, constructed with flood-resistant materials (if below the BFE), and designed or located where floodwater is prevented from entering or accumulating in the components that are not flood resistant.

What if my home is at a high point of a flood zone?

Flood maps are imperfect and changing. If your structure or land is on naturally high ground or filled land, there is an opportunity for one to request a map change. Any changes would also assist in a more accurate flood insurance rate. Such revisions will require an Elevation Certification.

Additional Links:

Additional Resources Available from DLNR:

[Floodplain Management Publications](#)

[Technical Bulletins](#)

[Guiding Regulations \(Federal / State / Local\)](#)

[Detached Solar Installations](#)

[Additional FAQs](#)

Additional Resources Available from DPP:

[Flood Zone Certifications & Checklists](#)

[Public contact directory omitted from this portfolio copy.](#)

Figures from the original document

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